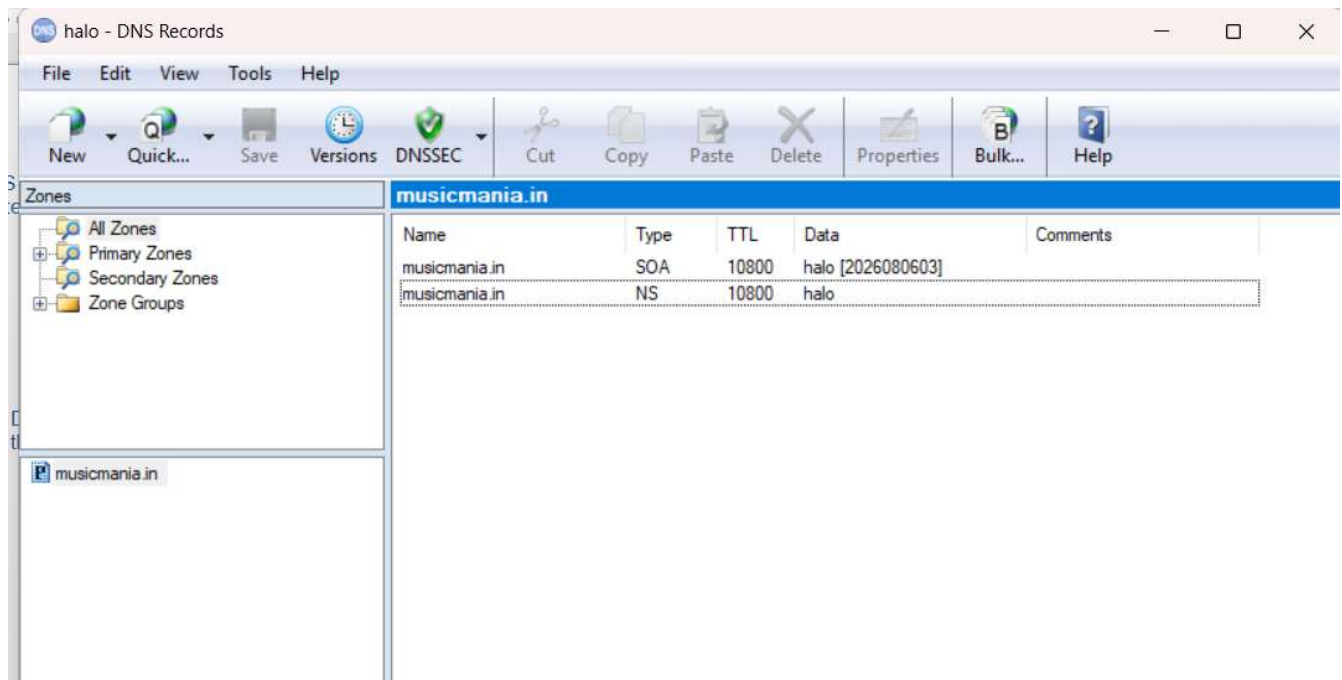
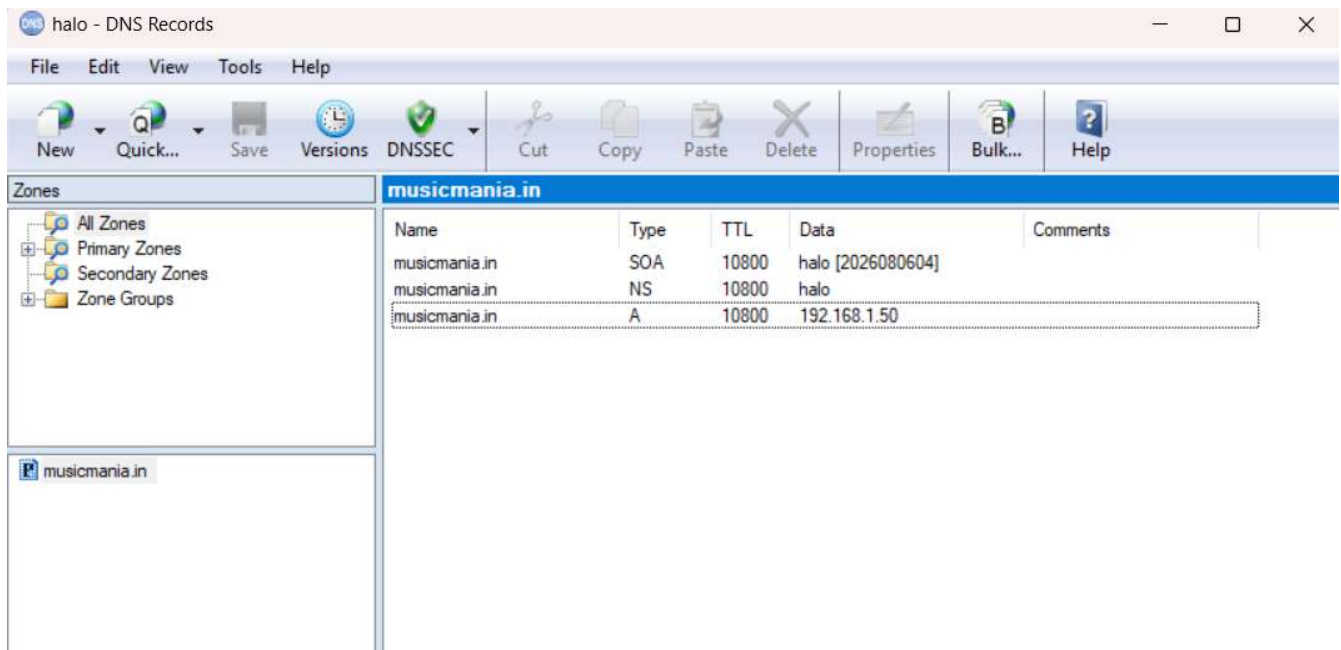


Day 3. DNS server Configuration

Task 1. Install and set up a DNS server on your local machine or a VM using BIND (Linux) or Simple DNS Plus (Windows), then create a new DNS zone for the domain musicmania.in.



Task 2. Add an A record in your DNS server configuration to map www.musicmania.in to the IP address 192.168.1.50, then use the nslookup or dig command to verify the mapping works.



Task 3. Simulate a typo by adding a CNAME record for play.musicmania.in that points to www.musicmania.in, then test if accessing play.musicmania.in resolves to the correct IP address.

```
C:\Windows\System32>NSLOOKUP PLAY.MUSICMANIA.IN 127.0.0.1
Server: localhost
Address: 127.0.0.1

Name:     MUSICMANIA.IN
Address:  192.168.1.50
Aliases:  PLAY.MUSICMANIA.IN

C:\Windows\System32>
```

- The output confirms that play.musicmania.in acts as an alias (CNAME) pointing to www.musicmania.in.
- www.musicmania.in subsequently resolves to the underlying target IP address (192.168.1.50).

Task 4. Your DNS server is not resolving the domain `musicmania.in` as expected. Document the step-by-step troubleshooting process you would follow, including checking configuration files, restarting the DNS service, and testing with `nslookup` or `dig`. Hint: Include at least three different troubleshooting commands or checks in your documentation.

Step 1: Verify the DNS Service Status

Check if the Simple DNS Plus background service is running and active:

- **Command:** Press `Win + R`, type `services.msc`, press **Enter**, and check the status of **Simple DNS Plus**.

	Sensor Data Service	Delivers data from a variety of sensors		Manual (Trigger Start)	Local System
	Sensor Monitoring Service	Monitors various sensors in order to exp...		Manual (Trigger Start)	Local Service
	Sensor Service	A service for sensors that manages differ...	Running	Manual (Trigger Start)	Local System
	Server	Supports file, print, and named-pipe sha...	Running	Automatic (Trigger Start)	Local System
	Shared PC Account Manager	Manages profiles and accounts on a Sha...		Disabled	Local System
	Shell Hardware Detection	Provides notifications for AutoPlay hard...	Running	Automatic	Local System
	Simple DNS Plus	Provides domain name services (DNS)		Automatic	Local System
	Smart Card	Manages access to smart cards read by t...		Manual (Trigger Start)	Local Service
	Smart Card Device Enumerat...	Creates software device nodes for all sm...		Manual (Trigger Start)	Local System
	Smart Card Removal Policy	Allows the system to be configured to lo...		Manual	Local System
	SNMP Trap	Receives trap messages generated by loc...		Manual	Local Service

Restart the service

	Sensor Data Service	Delivers data from a variety of sensors		Manual (Trigger Start)	Local System
	Sensor Monitoring Service	Monitors various sensors in order to exp...		Manual (Trigger Start)	Local Service
	Sensor Service	A service for sensors that manages differ...	Running	Manual (Trigger Start)	Local System
	Server	Supports file, print, and named-pipe sha...	Running	Automatic (Trigger Start)	Local System
	Shared PC Account Manager	Manages profiles and accounts on a Sha...		Disabled	Local System
	Shell Hardware Detection	Provides notifications for AutoPlay hard...	Running	Automatic	Local System
	Simple DNS Plus	Provides domain name services (DNS)	Running	Automatic	Local System
	Smart Card	Manages access to smart cards read by t...		Manual (Trigger Start)	Local Service
	Smart Card Device Enumerat...	Creates software device nodes for all sm...		Manual (Trigger Start)	Local System

Step 2: Audit Zone Configuration and Records

Inspect the zone properties in Simple DNS Plus for configuration errors:

1. Open **Simple DNS Plus** - click **DNS Records**.
2. Select **musicmania.in** under Primary Zones.
3. Verify that:
 - The **A Record** (`www`) is correctly pointing to the target IP (`192.168.1.50`).
 - The **CNAME Record** (`play`) points strictly to `www.musicmania.in`.
 - The zone status is enabled and no syntax errors or typos exist in the record names.

Step 3: Test Local Resolution with nslookup

Perform a direct local query to isolate whether the issue is with the DNS server itself or local host settings:

- **Command:** `nslookup www.musicmania.in 127.0.0.1`

```
C:\Users\srikantm05>nslookup www.musicmania.in 127.0.0.1
Server:    localhost
Address:   127.0.0.1

*** localhost can't find www.musicmania.in: Non-existent domain
```

Correct command is `nslookup musicmania.in 127.0.0.1`

```
C:\Users\srikantm05>nslookup musicmania.in 127.0.0.1.
Server:    localhost
Address:   127.0.0.1

Name:      musicmania.in
Address:   192.168.1.50
```

Analysis:

- **Success:** Returns 192.168.1.50, proving Simple DNS Plus is serving the zone properly.
- **Request timed out / No response:** Indicates the DNS service is not listening on port 53 or firewall is blocking the traffic.
- **Non-existent domain (NXDOMAIN):** Indicates the zone name is misspelled or not loaded properly in Simple DNS Plus.

Step 4: Flush Local DNS Resolver Cache and Verify Port Listening

If the server responds via 127.0.0.1 but your system still fails to resolve the domain, flush the Windows DNS cache and verify the service port:

- **Clear local cache:** `ipconfig /flushdns`
- **Verify port 53 binding:** `netstat -ano | findstr :53`